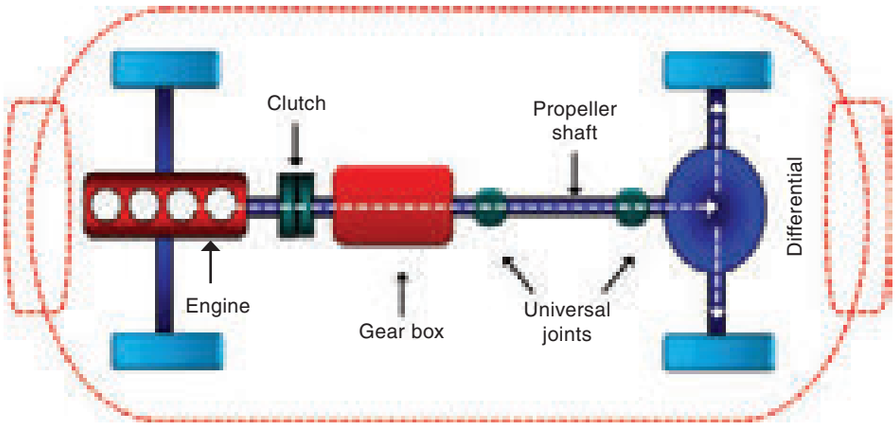


Figure 1.3.1 Front engine rear wheel drive

the running wheels is transmitted through the clutch, gearbox, universal joint and the differential unit. The front engine rear wheel drive is shown in figure 1.3.1.

Advantages

- Engine cooling is good
- Vehicle's weight is evenly distributed
- It is easy to connect the components with the engine
- It is easy to control the vehicle

Disadvantages

- Lengthy propeller shaft is needed
- Vehicle's ground clearance is high
- Vehicle's stability is low.

1.3.2 Front engine front wheel drive

In front engine front wheel drive, the clutch, gearbox and the differential unit are made as an integral part. The engine's power is transmitted to the front wheels by means of the clutch, gearbox, differential and small shaft. Propeller shaft is not necessary in such drives for transmitting the power.

Two pairs of constant velocity universal joints are used in front axle. The first one is located near the differential and another one is kept near the stub axle. To Transfer power with required angle and to transmit the steering power to the wheel these joints are being used. This type of drive is used in modern vehicles. Its construction is shown in figure 1.3.2.

Advantages

- Simple in design
- Have high stability
- Good road grip
- High engine cooling
- Good control over the vehicle
- There is no need of a separate propeller shaft

Disadvantages

- Maintenance cost is high
- Road grip is less when the vehicle is driven in slope or slanting regions
- Due to the engine's exhaust, passengers gets affected

1.3.3 Front engine four wheel drive

Four wheel drive is the one in which the engine's power is transmitted to all the four wheels at a time. The four

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